An Explicit Uniform Gap for the Lonely Runner Problem

Abstract

Let V be a set of $n \geq 2$ distinct positive integer speeds. We prove that there is a time t at which $\min_{v \in V} \|vt\| > (1 + e^{-6888})/(2n)$. The covering argument of Hu and Jing reduces the problem to a weighted greatest-common-divisor sum on separated speed sets. We control the distribution of prime valuations using a one-dimensional Green kernel, and combine a sieve for rough rational numbers with a dyadic decomposition to bound the sum. The finite numerical inequalities are certified by rational computations in a Lean formalization.

1 Introduction

Write $\|x\| = \min_{k \in \mathbb{Z}} |x - k|$ and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. For a finite set V of positive integer speeds, define

$$\mathrm{ML}(V) = \max_{t \in \mathbb{T}} \min_{v \in V} \|vt\|.$$

The lonely runner conjecture predicts that $\mathrm{ML}(V) \geq 1/(n+1)$ when $|V| = n$, corresponding to n moving runners and one stationary runner. Hu and Jing [1] proved that there is an absolute constant $c > 0$ for which $\mathrm{ML}(V) > (1 + c)/(2n)$ for every $n \geq 2$. We establish an explicit value of c.

Theorem 1. *If $V \subset \mathbb{Z}_{>0}$ is finite and $n = |V| \geq 2$, there exists $t \in [0, 1)$ such that*

$$|vt - k| > \frac{1 + e^{-6888}}{2n} \qquad (v \in V,\ k \in \mathbb{Z}).$$

The reduction uses the covering multiplicities of Hu and Jing and the valuation-layer approach to large common divisors of Green and Walker [2]. The rational certificates and their formalization are described in Appendix A.

For $0 \leq r \leq 1/2$ let $B_v(r) = \{t \in \mathbb{T} : \|vt\| \leq r\}$. Its normalized measure is $2r$. Set $r_0 = 1/(2n)$.

2 Circle coverings and spillover

2.1 Pairwise overlap

Following [1], we separate overlaps at a common center from those at distinct centers. Define $H(u, v) = \max(u, v)/\gcd(u, v)$. For $r < 1/2$, the arcs forming $B_v(r)$ have centers j/v and radius r/v, and are disjoint. The $\gcd(u, v)$ common centers contribute a total overlap $2r/H(u, v)$. Set

$$Q_r(u, v) = |B_u(r) \cap B_v(r)| - \frac{2r}{H(u, v)}.$$

Lemma 2. *For $0 \leq r < 1/2$,*

$$\max\{0, 4r^2 - 4r/H(u, v)\} \leq Q_r(u, v) \leq 4r^2.$$

Proof. Write $u = da$ and $v = db$, where $a \le b$ and $\gcd(a, b) = 1$. The map $t \mapsto dt$ preserves normalized measure, so it suffices to use the speeds a, b. The differences $j/a - k/b \bmod 1$ run once through the grid of mesh $1/(ab)$ by coprimality. On rescaling by ab, the two radii become $x = br$ and $h = ar$. Summing over lifts to the real line, the total overlap multiplied by ab is

$$\sum_{\ell \in \mathbb{Z}} |[-x, x] \cap [\ell - h, \ell + h]| = \int_{-h}^{h} \#\{\ell \in \mathbb{Z} : -x \le \ell + s \le x\}\, ds.$$

For $r > 0$, the integrand equals $\lfloor x - s \rfloor + \lfloor x + s \rfloor + 1$ almost everywhere. Writing $\{w\} = w - \lfloor w \rfloor$ and removing the contribution $2h$ of the common center gives

$$Q_r(u, v) = 4r^2 - \frac{2}{ab} \int_{x-h}^{x+h} \{w\}\, dw.$$

The bounds follow from $0 \le \{w\} < 1$ and the nonnegativity of the overlap at distinct centers. At $r = 0$, all arcs are singletons and the overlap has measure zero. □

2.2 Dilation of multiplicities

Let $N_r = \sum_{v \in V} \mathbf{1}_{B_v(r)}$, and write

$$E_k(r) = \int_{\mathbb{T}} (N_r - k)_+, \qquad g(V) = 1 - \left| \bigcup_{v \in V} B_v(r_0) \right|, \qquad x_+ = \max(x, 0).$$

Since $\int_{\mathbb{T}} N_{r_0} = 1$, we have $E_1(r_0) = g(V)$.

Lemma 3. *For a finite family of arcs dilated about their centers by $s \ge 1$, the integral of the positive part of the multiplicity minus k increases by at least a factor s, provided the dilated arcs have length less than 1. In particular, $E_k(sr) \ge sE_k(r)$ for every integer $k \ge 0$ when $sr < 1/2$.*

Proof. First consider intervals on the real line. With fixed radii ρ_i, the length of the union of the intervals centered at uc_i is a continuous piecewise affine function of $u > 0$. On an affine piece, if the extreme endpoints of a component come from intervals i, j, their extremality gives

$$u(c_i - c_j) \le \rho_i - \rho_j, \qquad u(c_i - c_j) \le \rho_j - \rho_i.$$

Thus the slope $c_j - c_i$ of the component length is nonnegative. Rescaling the line shows that the union length divided by the dilation factor is nonincreasing when the radii are dilated at fixed centers. This applies to every subfamily.

For $k \ge 1$, color the intervals in order of increasing left endpoint using k colors. Use an unused color if possible. Otherwise assign the color whose largest right endpoint so far is smallest. If a new interval and an earlier interval of the same color contain t, each other color already has an interval reaching t from the left. Consequently the number of colors at a point of multiplicity N is $\min(N, k)$. Any coloring has at most this many colors at a point. Therefore

$$\int (N - k)_+ = \sum_i |I_i| - \max_{\text{colorings}} \sum_{d=1}^{k} \left| \bigcup_{\text{color}(i)=d} I_i \right|.$$

The first term grows linearly with dilation, while the second term divided by the dilation factor is nonincreasing. For $k = 0$, the integral is the sum of interval lengths.

For the circle, choose one representative of each center in $[0, 1)$ and retain the translates indexed by $-M, \ldots, M$. The real-line integral differs from $2M + 1$ times the circle integral by a bounded endpoint error, since both the endpoint regions and the multiplicities there are uniformly bounded. Divide by $2M + 1$ and let $M \to \infty$. □

2.3 The spillover inequality

Sum over ordered pairs of distinct speeds and put

$$W(V) = \frac{1}{n^2} \sum_{u \neq v \in V} \min(1, 2n/H(u,v)), \qquad Q(r) = \sum_{u \neq v \in V} Q_r(u,v).$$

The difference between $\int N_r(N_r - 1)$ and $Q(r)$ is linear in r. The identity

$$m(m-1) - 2(m-1)_+ = 2\sum_{k=2}^{n-1}(m-k)_+ \quad (0 \le m \le n,\ m \in \mathbb{Z})$$

and Lemma 3 therefore give the second-moment spillover inequality of Hu and Jing [1]. For $s \ge 1$ and $0 \le r < 1/(2s)$,

$$Q(sr) \ge sQ(r) + 2(E_1(sr) - sE_1(r)).$$

Lemma 2 gives $Q(r_0) \ge 1 - 1/n - W(V)$ and $Q(sr_0) \le s^2(1 - 1/n)$. Also $E_1(sr_0) \ge s - 1$, since the sum of arc lengths is s and their union has measure at most 1. Substituting $s = 7/5$ and rearranging yields

$$g(V) \ge \frac{3}{35} - \frac{W(V)}{2} + \frac{1}{5n}. \tag{C1}$$

3 Reduction to separated speed sets

For $P > 0$, call a set P-separated if $H(u,v) > P$ for every pair of distinct elements. Suppose $W(S) \le \delta$ for all P-separated sets S, with $W(S)$ normalized by $|S|$ as above. Keep r_0 fixed and, for $S \subset V$, put

$$E(S) = \sum_{v \in S} |B_v(r_0)| - \left| \bigcup_{v \in S} B_v(r_0) \right|.$$

If $H(u,v) \le P$ for distinct $u, v \in S$, deleting u decreases $E(S)$ by at least the common-center overlap $1/(nP)$. Since $E(V) = g(V)$ and $E(S) \ge 0$, the nonempty separated set remaining at termination has $1 - z \le Pg(V)$, where $z = |S|/n$. For pairs within S, $\min(1, 2n/H) \le z^{-1}\min(1, 2|S|/H)$. The remaining pairs contribute at most $1 - z^2$ to $W(V)$. Since $W(S) = 0$ for a singleton, in all cases

$$W(V) \le zW(S) + 1 - z^2 \le \delta + 2Pg(V). \tag{C2}$$

We will use $P = e^{6880}$ and $\delta = 17/100$.

4 Stabilization of weighted graphs

Set $q = 201/50 = 4(1 + \sigma)$ and $\sigma = 1/200$. We use a bipartite graph with labeled positive integer vertices, with distinct values on each side, and with edge set E. If the side sizes are m_1, m_2, let $m = m_1 m_2 \le n^2$. For scales $a, b > 0$, set

$$w(x,y) = \min\left(1, \frac{\gcd(x,y)}{\max(ax, by)}\right), \quad \rho = \frac{\sum_{(x,y)\in E} w(x,y)}{m}, \quad J = \rho^q mab(m/n^2)^\sigma.$$

For the initial graph use two copies of V, all ordered pairs of distinct values, and $a = b = 1/(2n)$. Then $\rho = W(V)$ and $J = W(V)^q/4$.

Following the valuation-layer approach to GCD graphs [2, 1], allow restriction to a nonempty subrectangle. Also allow restriction to fixed p-adic valuation layers i, j on the

two sides, followed by division of their values by p^i, p^j and replacement of the scales by $(ap^{(i-j)_+}, bp^{(j-i)_+})$. The latter operation preserves every retained weight and multiplies ab by $p^{|i-j|}$. It also preserves same-side heights. An operation that strictly increases the score decreases the sum of all vertex values, by removing a vertex or reducing a value. Starting from positive edge mass, repeated improvement terminates at a state with score at least the initial score, in which no allowed operation increases the score.

At such a state, let α, β be the uniform vertex measures and let μ be the normalized edge weight. For a restriction to $I \times J'$, put $v = \alpha(I)\beta(J')$. When $v > 0$, the score ratio is $\mu(I \times J')^q v^{1+\sigma-q}$, so stability gives

$$\mu(I \times J') \le [\alpha(I)\beta(J')]^{3/4}. \tag{L04}$$

The inequality is immediate when $v = 0$. For a valuation-layer restriction the ratio has the additional factor $p^{|i-j|}$. Writing α_i, β_j for the valuation marginals and μ_p for the edge measure on valuation pairs, we obtain

$$\mu_p(i,j) \le p^{-|i-j|/q}(\alpha_i\beta_j)^{3/4}. \tag{L05}$$

Stabilizing the fourth-power score $\rho^4 mab$ separately gives (L04) and (L05) with 4 in place of q in the latter exponent, which is a stronger bound.

5 Bilinear bounds for the Green kernel

5.1 The continuous estimate

Let f be a nonnegative step function supported on finitely many unit cells $[i, i+1)$, $i \in \mathbb{Z}$. For $a > 0$, set

$$U(x) = \frac{1}{2a} \int_{\mathbb{R}} e^{-a|x-y|} f(y)\, dy.$$

Splitting the integral at $y = x$ shows that U is continuously differentiable and $U'' = a^2 U - f$ between the jumps of f. Both U and U' decay exponentially at infinity. Integration by parts, with cancellation of the terms at the jumps, gives $E := \int (U'^2 + a^2 U^2) = \int fU$. If f has positive integral, U attains a positive maximum M. Writing $L = \int U^4$, we have

$$E - \frac{a^2 L}{M^2} \ge 2a \int |U'| U \sqrt{1 - U^2/M^2}$$

by the nonnegativity of the square of $|U'| - aU\sqrt{1 - U^2/M^2}$. On each side of a maximum point, the total variation of $\Phi(U)$, where $\Phi(u) = \int_0^u s\sqrt{1 - s^2/M^2}\, ds$, is at least $\Phi(M) = M^2/3$. Thus

$$E \ge a^2 L/M^2 + 4aM^2/3 \ge \frac{4a^{3/2}}{\sqrt{3}} \sqrt{L}.$$

Hölder's inequality gives $E \le \|f\|_{4/3} L^{1/4}$, whence $E \le \sqrt{3}\, \|f\|_{4/3}^2/(4a^{3/2})$. For the potentials of f and g, the same integration by parts gives $\int (U_f' U_g' + a^2 U_f U_g) = \int f U_g$. The double integral below is $2a \int f U_g$. Cauchy–Schwarz for the mixed energy therefore gives

$$\iint e^{-a|x-y|} f(x) g(y)\, dx\, dy \le \frac{\sqrt{3}}{2\sqrt{a}} \|f\|_{4/3} \|g\|_{4/3}.$$

The bound holds also when either function vanishes almost everywhere.

5.2 Discretization and the diagonal term

Take $\lambda = e^{-a}$. For sequences x_i, y_j with finite support, use step functions constant on the unit intervals $[i, i+1)$, $[j, j+1)$. The integral of the exponential kernel over an off-diagonal cell is $A\lambda^{|i-j|}$ and over a diagonal cell is D_0, where

$$A = \frac{(1-\lambda)^2}{\lambda a^2}, \qquad D_0 = 2\int_0^1 (1-s)e^{-as}\,ds = \frac{2(a-1+\lambda)}{a^2}.$$

In particular, $A - D_0 = 2(\sinh a - a)/a^2 > 0$. The continuous estimate gives, for nonnegative sequences,

$$\sum_{i,j} \lambda^{|i-j|} x_i y_j \le C(\lambda)\|x\|_{4/3}\|y\|_{4/3} + d(\lambda)\sum_i x_i y_i,$$

where

$$C(\lambda) = \frac{\sqrt{3}}{2}\frac{\lambda a^{3/2}}{(1-\lambda)^2}, \qquad d(\lambda) = 1 - \frac{2\lambda(a-1+\lambda)}{(1-\lambda)^2}.$$

Since $\sum_i x_i y_i \le \|x\|_{4/3}\|y\|_{4/3}$, the bilinear norm is bounded by $\mathcal{N}(\lambda) := C(\lambda) + d(\lambda)$. For probability marginals α_i, β_i, the sequences $\alpha_i^{3/4}, \beta_i^{3/4}$ each have $\ell^{4/3}$ norm 1.

6 Concentration at large primes

6.1 Choosing the center and bootstrapping

Assume (L04) and (L05). Fix a prime $p > 142$ and put $\lambda = p^{-1/q}$. Summing (L05) gives a lower bound of 1 for the bilinear kernel sum. Since $p \ge 149$ and $\exp(62511/12500) < 149$, we may replace λ in this sum by the larger value $\lambda_0 = e^{-311/250}$. For $D = \sum_i (\alpha_i\beta_i)^{3/4}$, the diagonal estimate of Section 5 and the rational bounds in Appendix A yield

$$1 \le C(\lambda_0) + d(\lambda_0)D, \qquad C(\lambda_0) < 0.68365, \quad 0 < d(\lambda_0) < 0.39442.$$

At $D = 401/500$, the rational upper bound on the right is $1 - 629/25000000$, so $D > 401/500$. Let $m_* = \max_i(\alpha_i + \beta_i)/2$. Termwise, $(\alpha_i\beta_i)^{3/4} \le \sqrt{m_*}\sqrt{\alpha_i\beta_i}$. Cauchy–Schwarz gives $\sum_i \sqrt{\alpha_i\beta_i} \le 1$, hence $m_* \ge D^2$. Fix an index k attaining m_*, and write $z = 1 - (\alpha_k + \beta_k)/2$. Then $z \le 89199/250000 < 0.357$.

Return to the actual parameter λ while keeping k fixed. The center cell and the rectangle off both the center row and center column have masses at most $(1-z)^{3/2}$ and $z^{3/2}$, respectively. For the remainder of the row and column, use Hölder's inequality with $\sum_{i\ne k}\lambda^{4|i-k|} \le 2\lambda^4/(1-\lambda^4)$. A second application to the two marginals, together with $\alpha_k^3 + \beta_k^3 \le 1 + (1-2z)^3$, gives

$$z^{1/4} b(z) \le \frac{2\lambda}{(1-\lambda^4)^{1/4}}[1 + (1-2z)^3]^{1/4}, \qquad b(z) = \frac{1 - (1-z)^{3/2} - z^{3/2}}{z}.$$

The case $z = 0$ requires no further estimate. For $0 < z \le 1/2$, let $s = \sqrt{z} + \sqrt{1-z}$. Then $b(z) = 2(1-z)(s+2)/(s+1)^2$, so $b(z)/[1+(1-2z)^3]^{1/4}$ is positive and nonincreasing. Indeed, s is increasing and $(s+2)/(s+1)^2$ is decreasing. For the other factor $(1-z)/[1+(1-2z)^3]^{1/4}$, its fourth power is increasing in $t = 1-2z$, as $4 + t^3 - 3t^2 > 0$. Thus if $z \le S \le 1/2$,

$$z \le \frac{16\lambda^4[1 + (1-2S)^3]}{(1-\lambda^4)b(S)^4}.$$

Using $\lambda \le 144004613283/500000000000$ and the endpoints in Table 1 gives the successive upper bounds

$$0.357, \quad 0.342, \quad 0.312, \quad 0.262, \quad 0.201, \quad 0.150, \quad 0.121.$$

6.2 The exceptional edge mass at one prime

Let E_p be the mass of edges whose two valuations are both strictly below k or both strictly above k. Let z_-, z_+ be the averages, over the two marginals, of the masses strictly below and strictly above k, so $z = z_- + z_+$. For $z > 0$ put $u = \min(z_-, z_+)/z$, and define

$$f(u) = u^{3/4} + (1-u)^{3/4}, \quad h_0(u) = u^{3/2} + (1-u)^{3/2},$$
$$j(u) = [u(1-u)]^{3/4}, \qquad A_1 = (1-\lambda^4)^{-1/4}.$$

By (L04) and the arithmetic–geometric mean inequality, $E_p \le z^{3/2} h_0(u)$. For tails on opposite sides of k, the kernel factors into two geometric tails, each at distance at least one from k. Hölder bounds their combined mass by $A_1^2\lambda^2[(\alpha_-\beta_+)^{3/4} + (\alpha_+\beta_-)^{3/4}]$, where the subscripts denote the corresponding tails. This is at most $4^{3/4}A_1^2\lambda^2(z_-z_+)^{3/4}$, by Cauchy–Schwarz and $\sum t_i^{3/2} \le (\sum t_i)^{3/2}$ for nonnegative t_i. The four center-to-tail rectangles similarly have total mass at most $2^{3/4}A_1\lambda z^{3/4} f(u)[1 + (1-2z)^3]^{1/4}$; here the last factor bounds $(\alpha_k^3 + \beta_k^3)^{1/4}$. Adding the center cell accounts for all mass.

Write

$$A(z) = \frac{1-(1-z)^{3/2}}{z}, \qquad C_*(z) = [1 + (1-2z)^3]^{1/4}, \qquad h = h_0(u) + 4^{3/4}A_1^2\lambda^2 j(u).$$

The resulting inequality is

$$z^{1/4}[A(z) - h\sqrt{z}] \le 2^{3/4}A_1\lambda f(u) C_*(z).$$

For $0 < z \le 1/8$ and $\lambda \le 1/3$, one has $1/\sqrt{2} \le h \le 2$. The identity $A(z) = \frac{3}{2}\int_0^1 \sqrt{1-tz}\,dt$ gives $1 \le A \le 3/2$ and $A' \le -3/8$. Thus $A(z) - h\sqrt{z} > 0$, its derivative is at most $-11/8$, and its logarithmic derivative is at most $-11/12$. Also $C_*'/C_* \ge -3/4$, by $2t^2 \le 1 + t^3$ for $0 \le t \le 1$. The ratio $[A(z) - h\sqrt{z}]/C_*(z)$ is therefore nonincreasing. For $z \le S \le 1/8$ define

$$B(S, \lambda, u) = \frac{2^{3/4}A_1 f(u) C_*(S)}{A(S) - h\sqrt{S}}.$$

We obtain

$$z \le \lambda^4 B(S,\lambda,u)^4, \qquad E_p \le p^{-100/67} B(S,\lambda,u)^6 h_0(u),$$

using $6/q = 100/67$. If $z = 0$ then $E_p = 0$.

6.3 Summing the exceptional masses

Starting at $S = 0.121$, partition $[0, 1/2]$ into 256 intervals. On $[l, v]$, bound $f(u), h_0(u), j(u)$ above by $f(v), h_0(l), j(v)$, respectively, using their monotonicity on $[0, 1/2]$. In each prime range, use a rational upper bound λ_{U} for λ in B and retain the factor $p^{-100/67}$ in the bound for E_p. The certificate gives a positive lower bound d on the denominator, an upper bound b on the numerator, and verifies

$$b \le b_{\mathrm{U}} d, \qquad b^6 h_{0,\mathrm{U}} \le e_{\mathrm{U}} d^6, \qquad \lambda_{\mathrm{U}}^4 b_{\mathrm{U}}^4 \le S_{\mathrm{next}}.$$

Here $h_{0,\mathrm{U}}$ bounds h_0 on the cell. The rational enclosures of radicals are checked by integer powers. These tests give the next upper bound for z and the coefficient e_{U} for the exceptional mass throughout the cell.

The certified sums over $142 < p \le 10^4$, $10^4 < p \le 10^6$, and $p > 10^6$ are less than $21/50$, $23/1000$, and $1/125$, respectively. The first two ranges use finite sieve tables in which each discarded integer is accompanied by a proper divisor. On the first range the powers are

bounded individually; on the second, subinterval counts are multiplied by an upper bound for the summand on each subinterval. For the last range, primes are restricted to the eight reduced classes modulo 30, and each progression is bounded by its first term and an integral:

$$\sum_{j\ge 0}(b+30j)^{-s} \le b^{-s} + \frac{b^{1-s}}{30(s-1)}, \qquad s = 100/67.$$

The rational data for these checks are given in [3].

The total is less than $0.451 < 23/50$. Only finitely many primes occur among the vertex values; outside this set choose the center zero. By the union bound, a proportion at least $r_* := 27/50$ of the edge weight has the center between its endpoint valuations at every prime $p > 142$. For a graph of 142-rough integers, meaning integers with no prime divisor at most 142, use center zero also at all primes $p \le 142$.

7 Removing small prime factors

7.1 A cut estimate and a median inequality

Regard the two labeled sides as disjoint, and let $\omega = (\mu + \mu^{\mathsf{T}})/2$, with marginal π. For a vertex subset S, set $z = (\alpha(S) + \beta(S))/2$, extending each marginal by zero to the other side, and write $e = \mu(S, S^c) + \mu(S^c, S)$. By (L04),

$$e \ge 1 - z^{3/2} - (1-z)^{3/2}, \qquad \pi(S) \le z^{3/2} + e/2.$$

For $0 < z \le 1/2$, the first right-hand side is at least $(2^{3/2}-2)z^{3/2}$, by the monotonic decrease of $[1-(1-z)^{3/2}]/z^{3/2}$ and its value at $1/2$. The same bound holds at $z = 0$. It follows that $e \ge (2-\sqrt{2})\pi(S)$; for $z \ge 1/2$, use S^c. Hence

$$\omega(S, S^c) \ge (1 - 1/\sqrt{2}) \min\{\pi(S), \pi(S^c)\}.$$

For any real-valued vertex function F and a π-median m_F, integrate over the cuts $S = \{F > t\}$. The integral of the smaller side mass is $\mathbb{E}_\pi|F - m_F|$. The integral of the directed cut mass is $\frac{1}{2}\mathbb{E}_\mu|F(x) - F(y)|$, by symmetry of ω. Thus

$$\mathbb{E}_\pi|F - m_F| \le (1 + 1/\sqrt{2})\mathbb{E}_\mu|F(x) - F(y)|.$$

7.2 A weighted valuation moment

For each prime $p \le 142$, let k_p be a median of ν_p and put $B(x) = \sum_{p\le 142} |\nu_p(x) - k_p| \log p$. For $0 < \tau_p < 1/q$, (L05) and Section 5 give

$$\mathbb{E}_\mu e^{\tau_p|\nu_p(x)-\nu_p(y)|\log p} \le \mathcal{N}(p^{-1/q+\tau_p}).$$

By Jensen's inequality the expected edge distance $\mathbb{E}_\mu|\nu_p(x) - \nu_p(y)|\log p$ is at most $\log \mathcal{N}(p^{-1/q+\tau_p})/\tau_p$. The 34 choices in Table 2 give a sum less than 99.984727. Multiplying by the factor in Section 7.1 gives $\mathbb{E}_\pi B < \overline{D} := 170.685$.

Retain vertices with $B(x) \le 1375$. Markov's inequality bounds the discarded π-mass by $\overline{D}/1375$. The weight of edges retained at both ends is thus at least the fraction

$$\bar{t} = 1 - 2\overline{D}/1375 = \frac{103363}{137500}.$$

Both sides remain nonempty. If the area changes from m to m', the score is multiplied by at least $\bar{t}^q(m'/m)^{1+\sigma-q} \ge \bar{t}^q$. For the fourth-power score the corresponding factor is at least $\bar{t}^4$.

7.3 Stripping and restabilization

Write $x = s_x x_{\mathrm{r}}$, where all prime divisors of s_x are at most 142 and x_{r} is 142-rough. With $d_s = \gcd(s_x, s_y)$ one has

$$\gcd(x, y) = d_s \gcd(x_{\mathrm{r}}, y_{\mathrm{r}}), \qquad \max(ax, by) \geq d_s \max(ax_{\mathrm{r}}, by_{\mathrm{r}}).$$

Thus replacing each vertex by its rough part at fixed scales does not decrease the weights. For two vertices on the same side, comparison of the reduced numerators in H gives

$$H(x, y) \leq H(x_{\mathrm{r}}, y_{\mathrm{r}}) \exp\left(\sum_{p \leq 142} |\nu_p(x) - \nu_p(y)| \log p \right).$$

For retained vertices the exponent is at most $B(x) + B(y) \leq 2750$. If the original same-side heights exceed e^{6880}, the rough heights exceed e^{4130}. This also gives injectivity of stripping on each side. The rough graph can be stabilized again by Section 4, preserving roughness, separation, and the lower bound for its score.

8 Counting rough rationals and bounding a block

8.1 Rational height

For a rough graph, use the centers from Section 6 and put $N = \prod_p p^{k_p}$. For a vertex x write $x/N = a_x/b_x$ in lowest terms, and set $x' = a_x b_x$. On a retained edge (x, y), k_p lies between $\nu_p(x)$ and $\nu_p(y)$ for every prime. Consequently

$$x'y' = \frac{xy}{\gcd(x, y)^2},$$

since the exponent on the left is $|\nu_p(x) - k_p| + |\nu_p(y) - k_p| = |\nu_p(x) - \nu_p(y)|$. Define the height of a pair of fractions $a/b, c/d$ to be $H(ad, bc)$. Dividing both arguments by N leaves their relative height unchanged.

Lemma 4. *Let $R > 0$, $\ell \geq 4096$, and let $S \subset [R, 2R)$ be a finite e^ℓ-separated set of positive rationals with 142-rough numerators and denominators in lowest terms. For every $T > 0$,*

$$\#\{a/b \in S : ab \leq T\} \leq 1 + \frac{17}{25} \frac{T}{\ell},$$

where the fractions in the count are in lowest terms.

8.2 Counting numerators at a fixed denominator

We first reduce Lemma 4 to a pair of sieve estimates. Put $P = e^\ell$. If $\sqrt{2}T \leq P$, any two fractions in the count have $ad, bc < \sqrt{2}T$, since $(ad)(bc) \leq T^2$ and $1/2 < ad/(bc) < 2$. Separation thus allows at most one. We may assume $R \geq 1/\sqrt{2}$. Indeed, if $S \neq \varnothing$ and $R < 1/\sqrt{2}$, take reciprocals and use the least reciprocal as the new lower endpoint R'. Then $R' > 1/(2R) > 1/\sqrt{2}$, and all reciprocals lie in $[R', 2R')$. Roughness, the numerator-denominator product, and the relative height are invariant under reciprocation.

Fix a denominator $b \leq B := \sqrt{T/R}$. The numerators x lie in $[Rb, U_b]$, where $U_b = \min(2Rb, T/b)$. They are P-separated. If $U_b \leq P$, there is at most one. More generally take positive integers X, Z with $Z^2 \leq X$ and $XZ \leq P$, and factor each numerator as $x = sr$, with s supported on primes at most X and r X-rough, meaning that it has no prime divisor at most X.

If $r > U_b/X$, then $s < X$; the same r cannot occur for two numerators, since their height would be less than X. When $s = 1$ we have $r \in [Rb, U_b]$, and when $s > 1$ we have $s \geq 143$ and $r \leq U_b/143$. If $r \leq U_b/X$, choose a divisor $d \mid x$ with $1 \leq d \leq X$ and $(x/d)Z \leq U_b$. When $s \leq Z$, take $d = 1$, since $sZ \leq X$. Otherwise, if s has a prime factor greater than Z, take that factor. If it does not, multiply its prime factors, with multiplicity, until the product first exceeds Z; the product is at most $Z^2 \leq X$. In both cases $d > Z$ suffices. Assign to x the Z integers $(x/d)j$, $1 \leq j \leq Z$. A collision between the images of x, y would express x/y as a ratio of positive integers at most XZ, contradicting separation. Hence the number of numerators at denominator b is at most

$$N_X[Rb, U_b] + N_X[1, U_b/143] + U_b/Z,$$

where $N_X[I]$ counts the positive X-rough integers in I. An empty interval contributes zero.

8.3 A finite Selberg sieve

For an integer $X \geq 2$ and $0 < U \leq A$, we use

$$N_X[U, A] \leq \frac{A - U}{\log(X + 1)} + \frac{16X^2}{\log(X + 1)^2}.$$

To see this, put $G = \sum_{r \leq X,\ r \text{ squarefree}} 1/\varphi(r)$. With μ_{M} denoting the Möbius function, set

$$w_d = \frac{d}{G} \sum_{\substack{r \leq X,\ d \mid r \\ r \text{ squarefree}}} \frac{\mu_{\mathrm{M}}(r/d)\mu_{\mathrm{M}}(r)}{\varphi(r)}.$$

Then $w_1 = 1$, and the square of $\sum_{d \mid m} w_d$ equals 1 for X-rough m. Sum the square over the entire interval. The count of multiples of any least common multiple differs from $(A-U)/[d, e]$ by at most 1. The main quadratic form is

$$\sum_{d,e} \frac{w_d w_e}{[d, e]} = \sum_r \varphi(r) \left(\sum_{r \mid d} \frac{w_d}{d} \right)^2 = 1/G.$$

The first identity uses $\sum_{r \mid (d,e)} \varphi(r) = (d, e)$. Finite Möbius inversion evaluates the inner sum as $\mu_{\mathrm{M}}(r)/(G\varphi(r))$ for squarefree $r \leq X$ and as zero otherwise.

Grouping $\sum_{m \leq X} 1/m$ by the squarefree kernel $\prod_{p \mid m} p$ shows $G \geq \log(X + 1)$: the contribution of kernel r is at most $\prod_{p \mid r} (p - 1)^{-1} = 1/\varphi(r)$. For the error, the terms in the formula for w_d have a common sign for fixed d, so

$$\sum_d |w_d| = \frac{1}{G} \sum_{\substack{r \leq X \\ r \text{ squarefree}}} \prod_{p \mid r} \left(1 + \frac{2}{p - 1}\right) \leq \frac{X}{G} \prod_{p \leq X} \left(1 + \frac{2}{p(p - 1)}\right) \leq \frac{4X}{G}.$$

For the first inequality, expand the product after removing the squarefree restriction and bound the number of multiples of d by X/d. For the last inequality, the factor at 2 is 2; for odd primes $p = 2k + 1$, use $2/(p(p - 1)) \leq 2/[3k(k + 1)]$ and $e^{2/3} < 2$. The error is at most $(\sum_d |w_d|)^2$, as required.

8.4 Sieve parameters and summation over denominators

Use $X = \lfloor e^{\alpha \ell} \rfloor$ and $Z = \lfloor e^{\ell/10} \rfloor$. The choices $\alpha = 1/5, 4/5, 9/10$ satisfy the required packing conditions, and $1/Z \leq 2e^{-\ell/10}$, $\log(X + 1) > \alpha \ell$.

The case $B \geq 128$. Use $\alpha = 1/5$. Write $U(x) = \min(2Rx, T/x)$ and $G_1(x) = U(x) - Rx$. Both functions are unimodal on $(0, B]$, with maximum at $B/\sqrt{2}$. Rough denominators lie in the classes $1, 5 \bmod 6$. On a progression of mesh 6, the sum of either function is at most its integral divided by 6 plus its maximum. Indeed, remove the largest sampled term and compare each other term to the integral over the mesh interval toward that term. The two classes contribute an integral divided by 3 plus two maxima. The relevant integrals and maxima are

$$\int_0^B U = \frac{T(1+\log 2)}{2}, \qquad \int_0^B G_1 = \frac{T\log 2}{2}, \qquad \max U = \frac{\sqrt{2}T}{B}, \qquad \max G_1 = \frac{T}{\sqrt{2}B}.$$

The same comparison on the integer grid gives $\sum_b U_b \leq \int_0^B U + \max U < 3T/2$. The sieve error per denominator is at most $32e^{2\ell/5}/(\ell/5)^2$. Since there are at most B denominators and $B/T \leq \sqrt{2}e^{-\ell/2}$, the normalized sieve error, after multiplying by ℓ/T, is at most $(1200/\ell)e^{-\ell/10}$. The normalized packing error is at most $3\ell e^{-\ell/10}$. The normalized main term is at most

$$5\left[\frac{\log 2}{6} + \frac{2}{\sqrt{2}B} + \frac{1}{143}\left(\frac{1+\log 2}{6} + \frac{2\sqrt{2}}{B}\right)\right].$$

Using $B \geq 128$, $\log 2 < 0.7$, $1/\sqrt{2} \leq 3/4$, $\sqrt{2} \leq 3/2$, and $\ell \geq 4096$, the sum of the three bounds is less than $0.673 < 17/25$.

The case $B < 128$. Only 1 can occur as a rough denominator. Write $A = \min(2R, T)$. If $A \leq e^{\ell}$, the count is at most one. If $A > e^{1.7\ell}$, use $\alpha = 4/5$ and $A - R \leq A/2$. The sieve and packing bounds give at most

$$\frac{A}{\ell}\left[\frac{1/2 + 1/143}{4/5} + (50/\ell + 2\ell)e^{-\ell/10}\right] < 0.635\, A/\ell.$$

If $e^{\ell} < A \leq e^{1.7\ell}$, choose $\alpha = 9/10$, so $A < X^2$ and each X-rough integer at most A is 1 or prime. For $\vartheta(x) = \sum_{p \leq x} \log p$ we use

$$0.92x \leq \vartheta(x) \leq 1.107x \qquad (\log x \geq 1000). \tag{1}$$

The first rough interval has count at most $(\vartheta(A) - \vartheta(R-1))/(\ell - 1)$, using $R - 1$ to retain the left endpoint. The second has count at most $1 + \vartheta(A/143)/(0.9\ell - 1)$. With $R \geq A/2$, the combined count including the packing term is less than

$$1 + \frac{0.647A + 0.92}{\ell - 1} + \frac{1.107A}{143(0.9\ell - 1)} + 2Ae^{-\ell/10} < 1 + 0.657\, A/\ell.$$

Since $A \leq T$, all cases give Lemma 4. □

Proof of (1). Write $F(x) = \log(\lfloor x \rfloor!)$ and $J(x) = F(x) - F(x/2) - F(x/3) - F(x/5) + F(x/30)$. The prime-power expansion of the factorial gives the coefficient $j(\lfloor x/p^k \rfloor)$ of the contribution from p^k, where $j(n) = n - \lfloor n/2 \rfloor - \lfloor n/3 \rfloor - \lfloor n/5 \rfloor + \lfloor n/30 \rfloor$. Checking residues modulo 30 gives $j \in \{0, 1\}$, with $j(n) = 1$ for $1 \leq n < 6$. Thus $J(x) \leq \psi(x) \leq J(x) + \psi(x/6)$, where ψ sums over prime powers. Integral comparison for F gives

$$|J(x) - C_0 x| \leq 5 + 10\log x \quad (x \geq 30), \qquad C_0 = \frac{\log 2}{2} + \frac{\log 3}{3} + \frac{\log 5}{5} - \frac{\log 30}{30}.$$

For instance, $|F(x) - (x\log x - x)| \leq 1 + 2\log x$ for $x \geq 1$, by comparison with the integral of log and replacement of $\lfloor x \rfloor$ by x. The upper recurrence gives $\psi(x) \leq (6/5)C_0 x + 200(1 + \log(x+1))^2$ by strong induction on $\lfloor x \rfloor$. For $x < 30$ the error term alone suffices. For $x \geq 30$, the difference of the error terms at x and $x/6$ exceeds $5 + 10\log x$. The prime-power contribution $\psi(x) - \vartheta(x)$ is at most $\sqrt{x}\log x$, since each prime at most $\sqrt{x}$ contributes at most $\log x$. These estimates at $\log x \geq 1000$ and rational enclosures for C_0 give (1).

8.5 A truncated double sum

Let $x'_1, \dots, x'_a$ and $y'_1, \dots, y'_b$ be nonempty lists of positive numbers with counting functions at most $1 + \kappa T$ for every $T > 0$. Then for $c > 0$,

$$\sum_{i,j} \min(1, c/\sqrt{x'_i y'_j}) \le a + b + ab\,\Psi\left(\frac{c\kappa}{\sqrt{ab}}\right), \qquad \Psi(t) = \begin{cases} 4t - 3t^2 + 2t^2 \log t, & 0 < t \le 1, \\ 1, & t \ge 1. \end{cases}$$

Order each list and remove its first entry. Each remaining i-th entry is at least i/κ. The removed row and column cost at most $a + b$; each remaining summand is bounded by the integral of the corresponding decreasing function on its unit cell. Rescaling the rectangle gives $\Psi(t) = \int_{[0,1]^2} \min(1, t/\sqrt{uv})\, du\, dv$. For $t \le 1$, integrating first in v and splitting at $u = t^2$ gives contributions t^2 and $4t(1-t) + 2t^2 \log t$. If both counting functions are at most κT, the same argument needs no deletion and the term $a + b$ may be omitted. The lists may contain repeated values, counted with their multiplicities.

Set $\varepsilon = 1/1000$, $t_0 = 19/100$, and $r_* = 27/50$. For $t > 0$ and $0 \le \rho \le 1$, the condition $r_*(1-\varepsilon)\rho \le \Psi(t)$ implies $\rho^4/t^2 \le 1/t_0^2$. For $t \ge t_0$ this follows from $\rho \le 1$. For $0 < t \le t_0$, we have $\log t \le -\log 5 < -8/5$, so $\Psi(t) \le F(t)$, where $F(t) = 4t - (31/5)t^2$. Writing $u = \sqrt{t/t_0} \in (0, 1]$, we obtain

$$uF(t_0) - F(t) = u(1-u)\left(4t_0 - \frac{31}{5} t_0^2 (1 + u + u^2)\right) \ge 0.$$

Here $1 + u + u^2 \le 3$. Since $F(t_0) = 0.53618 < r_*(1-\varepsilon)$, we get $\rho \le u$.

8.6 A block estimate at fixed cutoff

Consider a graph whose sides lie in $[A, 2A)$ and $[B, 2B)$, each consisting of distinct 142-rough integers, with side sizes $a, b \le n$. At cutoff $D > 0$, give an edge weight $\min(1, \gcd(x, y)/D)$, and let ρ be the sum of edge weights divided by ab. Put $Q_0 = AB/D^2$ and

$$J_{\mathrm{ph}} = \rho^q \frac{ab}{Q_0} (ab/n^2)^\sigma.$$

Upon restricting to valuation layers i, j at p and dividing by p^i, p^j, replace A, B, D by $A/p^i, B/p^j, D/p^{\min(i,j)}$. The retained weights are unchanged and Q_0 is multiplied by $p^{-|i-j|}$. Thus the stabilization argument applies to J_{ph} as well as, separately, to $\rho^4 ab/Q_0$, yielding (L04) and (L05).

In a stable graph of positive weight, retain the edges specified in Section 6, of mass at least $r_* \rho ab$. For their common normalization N, the identity of Section 8.1 and $xy < 4AB$ bound the retained weight by $\min(1, 2\sqrt{Q_0}/\sqrt{x'y'})$. Summing this upper bound over the entire labeled rectangle is permitted.

A fourth-power bound without separation. For reduced positive fractions in $[R, 2R)$ with numerator-denominator product at most T, the numerator is at most $\sqrt{2RT}$ and the denominator at most $\sqrt{T/R}$. There are at most $\sqrt{2}T$ such integer pairs. In the stable graph for the fourth-power score, apply the double-sum bound without deletion at $\kappa = \sqrt{2}$. It gives $r_*\rho \le \Psi(2\sqrt{2Q_0/(ab)})$ and hence $\rho^4 ab/Q_0 \le 8/t_0^2$. Since stabilization does not decrease the score, this holds for every original block.

The q-power score with separation. Assume both sides are e^{ℓ}-separated and $\ell \geq 4096$. In the stable graph for J_{ph} use Lemma 4 with $\kappa = (17/25)/\ell$. If $\min(\rho a, \rho b) \geq 2/(r_*\varepsilon)$, then $a + b \leq r_*\varepsilon\rho ab$. Subtracting this term from the double-sum bound yields $r_*(1-\varepsilon)\rho \leq \Psi(2\kappa\sqrt{Q_0/(ab)})$. Thus $\rho^4 ab/Q_0 \leq 4\kappa^2/t_0^2$, which also bounds J_{ph}. If $\rho a < 2/(r_*\varepsilon)$, then $\rho^4 ab/n^2 \leq 2/(r_*\varepsilon n)$ by $b \leq n$ and $\rho^3 \leq 1$, and the case of small ρb is symmetric. As $q = 4(1+\sigma)$, the score equals the fourth-power score times $(\rho^4 ab/n^2)^{\sigma}$. Use the fourth-power bound in this case. Combining the two bounds, and including zero weight, gives for the original block

$$J_{\mathrm{ph}} \leq C_{\mathrm{ph}}(\ell, n) := \frac{4}{t_0^2}\left(\frac{17}{25\ell}\right)^2 + \frac{8}{t_0^2}\left(\frac{2}{r_*\varepsilon n}\right)^{\sigma}. \tag{2}$$

9 Global estimate and proof of the theorem

9.1 Dyadic decomposition

In the rough graph of Section 7, partition ax and by using the common grid $[h2^i, h2^{i+1})$, $i \geq 0$, with $h = \min(a, b)$. Both scaled speeds are at least h. Put $A_i = h2^i/a$, $B_j = h2^j/b$, and $D_{ij} = \max(aA_i, bB_j)$. On this block the original weight is bounded by the fixed-cutoff weight at D_{ij}, and

$$\frac{A_i B_j}{D_{ij}^2} = \frac{1}{ab}\exp(-|\log(aA_i) - \log(bB_j)|).$$

Write m_i, n_j for the respective block sizes and m_L, m_R for the full side sizes. Taking a q-th root in (2) bounds the block's total weight by

$$C_{\mathrm{ph}}(\ell, n)^{1/q}(ab)^{-1/q} n^{2\sigma/q}(m_i n_j)^{3/4}(2^{-1/q})^{|i-j|},$$

because $1-(1+\sigma)/q = 3/4$. The sequences $(m_i/m_L)^{3/4}$ and $(n_j/m_R)^{3/4}$ have unit $\ell^{4/3}$ norms. Applying Section 5 to the sum, and returning to the definition of J, gives

$$J \leq \overline{B}^{\,q} C_{\mathrm{ph}}(\ell, n), \qquad \mathcal{N}(2^{-1/q}) \leq \overline{B} := 2.137859.$$

9.2 The bound for separated sets

Suppose V is e^{6880}-separated. If $n < e^{6880}/40$, then every ordered pair contributes a summand less than $1/20$ in $W(V)$, so $W(V) < 1/20$. Otherwise, the initial score, the retention of at least the fraction $\bar{t}^q$ of the score, and the resulting separation parameter $\ell = 4130$ give

$$\frac{\bar{t}^q W(V)^q}{4} \leq \overline{B}^{\,q} C_{\mathrm{ph}}(4130, n).$$

Since $C_{\mathrm{ph}}(4130, n)$ is decreasing in n, use $n \geq e^{6880}/40$. The certificates give

$$C_{\mathrm{ph}}(4130, e^{6880}/40) < \overline{C} := \frac{3004}{10^9}, \qquad 4\overline{C}\,\overline{B}^{\,q} < (0.1699\bar{t})^q.$$

Raising the second comparison to the fiftieth power reduces it to a rational inequality. It follows that $W(V) < 0.1699 < 17/100$.

9.3 Increasing the forbidden radius

Proof of Theorem 1. Apply (C2) with $P = e^{6880}$ and $\delta = 17/100$, and combine it with (C1) to obtain

$$(1 + e^{6880})g(V) \geq \frac{3}{35} - \frac{17}{200} = \frac{1}{1400}.$$

Since $\sum_{j=0}^{20} 8^j/j! > 2800$, we have $e^8 > 2800$ and $1400(1+e^{6880}) < e^{6888}$. Thus $g(V) > e^{-6888}$.

Set $c = e^{-6888}$ and increase the radius from r_0 to $(1+c)/(2n)$. For each speed the forbidden measure increases by c/n, so the total increase in the measure of the union is at most c. Its complement still has positive measure. Any point in the complement, represented in $[0, 1)$, lies outside every closed forbidden set and satisfies the strict inequalities of the theorem. □

A Numerical certificates and formalization

A.1 Rational data

For each row of Table 1, the rational numbers u, v are upper bounds on $\sqrt{S}, (1 - S)^{3/2}$, respectively, so that $(1 - v)/S - u$ is a lower bound on $b(S)$. The output T is an upper bound on the next value of z.

Table 1: Rational endpoint certificates for the six iterations.

Input S	Output T	Upper bound u	Upper bound v
357/1000	171/500	5974947699/10000000000	1031208431/2000000000
171/500	39/125	5848076607/10000000000	1067502341/2000000000
39/125	131/500	2792848009/5000000000	5706668661/10000000000
131/500	201/1000	5118593557/10000000000	6339931167/10000000000
201/1000	3/20	896660471/2000000000	3571002657/5000000000
3/20	121/1000	3872983347/10000000000	7836612789/10000000000

Table 2 gives the tilting parameters and certified upper bounds on $\log \mathcal{N}(p^{-1/q+\tau_p})/\tau_p$.

Table 2: Tilts and weighted valuation distances at small primes.

Prime p	Tilt τ_p	Rational upper bound
2	37 / 201	7670990696489 / 1000000000000
3	347 / 2010	640388639331 / 100000000000
5	107 / 670	662523832733 / 125000000000
7	61 / 402	2365565222337 / 500000000000
11	142 / 1005	4090203987259 / 1000000000000
13	277 / 2010	155231431817 / 40000000000
17	133 / 1005	3569839071111 / 1000000000000
19	131 / 1005	1724576903557 / 500000000000
23	17 / 134	3252068305893 / 1000000000000
29	41 / 335	151444678617 / 50000000000
31	122 / 1005	2967702979251 / 1000000000000
37	119 / 1005	2811502620707 / 1000000000000
41	39 / 335	681198415611 / 250000000000
43	233 / 2010	671373512687 / 250000000000
47	23 / 201	1306830035577 / 500000000000
53	113 / 1005	314968774063 / 125000000000
59	223 / 2010	2438837267921 / 1000000000000
61	37 / 335	603558459043 / 250000000000
67	22 / 201	117317759673 / 50000000000
71	109 / 1005	230537391869 / 100000000000
73	217 / 2010	285750766663 / 125000000000
79	43 / 402	223184591259 / 100000000000
83	107 / 1005	2198656928103 / 1000000000000
89	106 / 1005	2152619194079 / 1000000000000
97	7 / 67	1048611830883 / 500000000000

Prime p	Tilt τ_p	Rational upper bound
101	209 / 2010	258966201167 / 125000000000
103	104 / 1005	1029736157291 / 500000000000
107	69 / 670	2035878906713 / 1000000000000
109	69 / 670	2024512845143 / 1000000000000
113	103 / 1005	2002582458967 / 1000000000000
127	203 / 2010	483303801639 / 250000000000
131	101 / 1005	957616588461 / 500000000000
137	1 / 10	188956832491 / 100000000000
139	1 / 10	734896491 / 390625000

A.2 Verification and source locations

The rational power tests reduce to integer powers. For example, for positive integers a, b, n, the inequality $a/b \geq n^{-100/67}$ is equivalent to $a^{67}n^{100} \geq b^{67}$. Logarithms are enclosed by truncating

$$\log x = 2\sum_{j\geq 0} \frac{v^{2j+1}}{2j+1}, \qquad v = \frac{x-1}{x+1},$$

and bounding the remainder by a geometric series. The large-prime interval certificates and sieve tables are supplied with the Lean material [3]. Module names in Table 3 refer to files under `LonelyRunnerNew/`. The main theorem is the declaration `LonelyRunnerNew.lonely_runner_gap` in `SpilloverInputProved.lean`.

Table 3: Proof steps and corresponding Lean modules.

Step	Module or declaration
Covering and deletion	`TwoSpeedIntersection`, `PeriodicLevelDilation`, `SpilloverInputProved`, `ForestInputProved`
Stabilization and kernel	`InitialTaggedState`, `ValuationDistribution`, `GreenKernel`, `KernelInputProved`
Large-prime exceptions	`FixedScaleNumerics`, `BootstrapChain`, `LCNumericChecker`, `LCFiniteBudgets`, `LCQuantitativeRetention`
Removing small factors	`SmallPrimeRowsProved`, `CoreCut`, `JointCore`, `OriginalSeparatedPeeling`
Sieve and block bound	`FiniteIntervalSieve`, `ChebyshevExplicit`, `RationalCountProved`, `SaturatedTerminal`, `RoughBlockProved`
Global bound and theorem	`DyadicPhysicalPartition`, `DyadicKernelCertificate`, `DyadicRoughAssembly`, `NumericEndgame`, `SpilloverInputProved`

References

[1] X. Hu and Y. Jing, *Near optimal gap in the Lonely Runner Conjecture*, arXiv:2610.23790v1, 2026.

[2] B. Green and A. Walker, *Extremal problems for GCDs*, Combin. Probab. Comput. **30** (2021), 922–929. arXiv:2012.02078.

[3] *LonelyRunner_New*, Lean formalization and rational certificates, source manifest identifier `67008acda1805ce8a67e8557e793b7c91e6d016634d72da04dacc50ace443a9a`.